

MAYUR VEER

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EDUCATION

Northeastern University, Boston, MA

Expected May 2026

Master of Science in Software Engineering Systems

Relevant courses: Concepts of Object-Oriented Design, Web Design and User Experience Engineering, Program Structure and Algorithms, Network Structures and Cloud Computing, High Performance Computing for AI, User Experience Design and Testing

PROFESSIONAL EXPERIENCE

Ernst & Young LLP, Mumbai, INDIA

Sept 2021 – Aug 2024

Team Lead – Software Engineer

- Optimized performance and developed persistent state management, reducing unnecessary re-renders and load times by 60% by implementing React's **useCallback** hook for memoized functions, context selectors for targeted state updates, and **localStorage** for data persistence
- Reduced rendering time by 50% by using Angular's lifecycle hook **ngOnChanges** to refresh data based on conditions instead of reloading the whole page
- Achieved reduction in manual effort by 90% building an Arabic Invoice Generator for Formula 1, automating the process of reading English PDFs translating them to Arabic with **context-aware** accuracy translation library, and dynamically generating **RTL-formatted** invoices with precise text alignment and layout
- Cut execution time by 66.7% designing a recursive Python script to convert JSON files to tabular data and flat files to back to **JSON** in the same format, while effectively managing exceptional scenarios
- Optimized processing efficiency by 17.14x and reduced processing time by 94.17% refining code using **Pandas** that converts flat files to a tree structure, decreasing the time from 1 hour for 1,800 invoices to just 3.5 minutes for 25,000 invoices

PROJECTS

End-to-End Cloud Integration & CI/CD Automation | Terraform, Shell, Packer, AWS, GCP, Python, GitHub Actions

- Automated infrastructure provisioning for 3+ environments by creating 15+ resources (VPCs, EC2/VMs, ALBs, ASGs, target groups) using **Terraform across AWS and GCP**, reducing manual effort by 80%
- Improved subdomain reliability by 99.9% by configuring **Route53 via Terraform** to handle DNS routing for dev and demo, and attaching **SSL** from ACM/ssls.com to respective load balancers with health checks and multi-AZ distribution for high availability
- Enhanced infrastructure security for 100% of services by integrating **AWS KMS** encryption (EC2, RDS, S3), **Secrets Manager** for credentials, and **CloudWatch** for real-time logging and alerts, with autoscaling groups configured for automatic security patching and compliance
- Reduced image provisioning time by 60% by automating **AMI and GCP** image creation with **Packer and shell scripts** for consistent deployment environments, optimized for autoscaling launch templates with pre-configured monitoring agents
- Decreased deployment time by 75% by building **CI/CD** pipelines using **GitHub Actions** with **PyTest** for automated API testing, enabling seamless rollout of a Python-PostgreSQL API via instance refresh and AMI updates with zero-downtime deployments through Application Load Balancer target group management and rolling updates

Eventify | TypeScript, MongoDB, Express, React, NodeJS, HTML, SCSS, Git

- Built a full-stack event platform with 5+ core modules (auth, booking, search, payments, chat), improving user retention by 30% using **React Redux**, PWA offline support
- Implemented **Internationalization (i18n)** supporting 6+ languages, expanding accessibility across global regions
- Boosted payment processing efficiency by 40% by integrating **Stripe**, **Google Maps**, and QR-based check-ins, streamlining event logistics for 100+ users
- Increased user engagement by 25% by implementing **Fugu API for real-time** chat between event organizers and attendees, enhancing support and communication

Sentiment Aura | Python, FastAPI, React, Claude API, Deepgram, P5.js, Docker, GCP, WebSocket

- Built real-time AI-powered emotion visualization app that converts speech to dynamic weather patterns with <600ms latency
- Integrated Deepgram speech-to-text and Claude sentiment analysis APIs with WebSocket streaming for **live audio processing**
- Created procedural P5.js **weather visualization** with multi-layer clouds, and smooth transitions between 5 weather states
- Containerized and deployed** full-stack application on Google Cloud Platform with automated build pipeline

Netflix Clone | JavaScript, MongoDB, Express, React, NodeJS, HTML, CSS, Git

- Developed a Netflix-style streaming platform with responsive UI and dynamic landing page using **Tailwind CSS**, resulting in a 50% increase in session time across devices
- Enhanced content discovery by 40% by integrating **TMDB API** for real-time trailers and previews of trending media
- Secured user sessions with 100% **JWT-based** authentication system, reducing unauthorized access and session errors by 90%

Data Processor Backend | Python, FastAPI, GCP Cloud Run, Pub/Sub, Firestore, Terraform, Docker, GitHub Actions

- Built high-throughput data ingestion API processing **2,133 requests/minute with 1,169ms average response time under 1,000 concurrent requests**, using FastAPI and serverless Cloud Run infrastructure
- Implemented resilient async processing pipeline with Pub/Sub message queue, achieving **95% success rate and zero message loss** through automatic NACK-based retries and exponential backoff strategy
- Designed multi-tenant security architecture using Firestore document sub-collections ensuring physical data isolation, with **PII redaction** for compliance and tenant-scoped queries
- Automated full deployment lifecycle with Terraform managing **15+ GCP resources** and GitHub Actions CI/CD featuring **21 PyTest unit tests**, code quality checks (Black, flake8, isort), and end-to-end integration validation

TECHNICAL SKILLS

- Languages:** Python, Java, JavaScript, TypeScript, HTML, CSS, SCSS, SQL, Shell, YAML, HCL
- Frameworks:** Flask, FastAPI, Angular | **Databases:** MSSQL, PostgreSQL, SQLite | **Tools:** Power BI, Tableau
- DevOps:** CI/CD pipeline, Kubernetes, Docker, GitHub Actions, Terraform, Packer, AWS, GCP, Linux
- Certificates:** Data Science and Statistics, Deep Learning, R Programming, Tableau – Internshala (2021), Python – Sadguru classes (2020), Machine Learning with Python – K.J. Somaiya College of Engineering (2020)